

Data Science

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Extracting insight and value from data

Data Science combines statistics, programming, and domain expertise to extract meaningful insights and predictions from structured and unstructured data.

DEFINITION

An interdisciplinary field that uses scientific methods, algorithms, and systems to extract knowledge and insights from structured and unstructured data.

WHY IT MATTERS

Organizations generate massive amounts of data — data science turns that raw data into actionable business and scientific insight.

WORKING PRINCIPLE



Data scientists collect and clean data, apply statistical and machine learning methods to identify patterns, and communicate findings through visualizations and models that support decision-making.

QUICK FACTS

- ▶ It's estimated data scientists spend up to 80% of their time cleaning and preparing data before analysis
- ▶ The world generates an estimated 2.5 quintillion bytes of data every day

KEY TERMS

- Dataset
- Statistical model
- Data visualization
- Predictive analytics

CORE CONCEPTS

- ▶ Data cleaning and preprocessing
- ▶ Statistical analysis
- ▶ Data visualization
- ▶ Predictive modeling

APPLICATIONS

- ▶ Business trend analysis
- ▶ Predictive customer analytics
- ▶ A/B testing and experimentation
- ▶ Scientific data analysis

REAL-LIFE EXAMPLES

- ▶ E-commerce sites predicting customer purchases
- ▶ Sports teams using data analytics for strategy
- ▶ Public health data tracking disease trends

ADVANTAGES

- ▶ Enables data-driven decision-making
- ▶ Uncovers patterns invisible to manual analysis
- ▶ Broadly applicable across nearly every industry

LIMITATIONS

- ▶ Poor quality data leads to poor conclusions
- ▶ Requires diverse skills across statistics, coding, and domain knowledge
- ▶ Correlation found in data doesn't always mean causation

CAREER OPPORTUNITIES

Data scientist, Data analyst, Business intelligence analyst, Data engineer

INDUSTRY APPLICATIONS

Technology, Finance, Retail, Healthcare

RESEARCH AREAS

Automated data analysis (AutoML), Causal inference methods, Big data infrastructure

FUTURE SCOPE

Growing data volumes and AI integration are expected to keep data science among the highest-demand career fields.

CURRENT TRENDS

Increasing use of AI-assisted tools that automate parts of the data cleaning and analysis workflow.

SUMMARY

Data science turns raw data into actionable insight using statistics, programming, and visualization, powering decisions across every industry.